A two-dimensional reaction-advection-diffusion **model** of the **spread** of fire in wildlands

P Grasso, MS Innocente - 2018 - ap1.sib.uc.pt

… The aim of this research is to develop a **model** of the **spread** of wildfires … The **model** consists of two coupled partial differential … The aim of this research is to **model** the **spread** of **forest fire** …

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Atmospheric interactions with **wildland fire** behaviour–I. Basic surface interactions, vertical profiles and synoptic structures

BE Potter - International Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… rate and wind speed based on the results of their numerical **model** with coupled fire … Part II of this **review** discusses plume structure and **dynamics** in detail, including elaboration on …

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Estimation of **wildfire** size and risk changes due to fuels treatments

MA Cochrane, CJ Moran, MC Wimberly… - … of **Wildland Fire**, 2012 - CSIRO Publishing

… modelling system (Finney 2004) to simulate the observed **wildfire** progression and **spread** rates. **Model** simulations were parameterised with LANDFIRE fuels and topographic data …

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Development and structure of Prometheus: the Canadian **wildland fire** growth **simulation model**

C Tymstra, RW Bryce, BM Wotton, SW Taylor… - … Report NOR-X-417 …, 2010 - spyd.com

… • to **review** and evaluate existing models; • to … a **wildfire** at many points around the fire perimeter, including the underlying rates of **spread** for the spatial **simulation** of firefront **propagation**. …

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Distributed **wildfire** surveillance with autonomous aircraft using deep reinforcement learning

KD Julian, MJ Kochenderfer - … of Guidance, Control, and **Dynamics**, 2019 - arc.aiaa.org

… work together to monitor the **wildfire** growth. Furthermore, the … for both **wildfire propagation** and **trajectory** optimization is … **simulation**, the **wildfire** is small, and so no precise **trajectory** …

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A **review** of radiant heat flux models used in **bushfire** applications

AL Sullivan, PF Ellis, IK Knight - … Journal of **Wildland Fire**, 2003 - CSIRO Publishing

… The **prediction** of the thermal radiation from a **wildfire** is … is important in a number of fields, including rate of **spread** modelling, … of heat transfer and fire **propagation**, the physics is reason…

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Federal **forest-fire** policy in the United States

SL Stephens, LW Ruth - Ecological applications, 2005 - Wiley Online Library

… **wildfire** has increased over the last decade. Given the current and future challenges of fire **management**, and based on analytical research and **review** … not contribute to **wildfire spread** or …

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What do we know about **forest fire** size distribution, and why is this knowledge useful for forest **management**?

W Cui, AH Perera - International Journal of **Wildland Fire**, 2008 - CSIRO Publishing

… in forest **management** and **forest fire management**. In the present article, our goal is to **review** and … For example, slope affects rate of **spread** while aspect affects **spread** direction. …

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Analysis of evacuation **behavior** in a **wildfire** event

T Toledo, I Marom, E Grimberg, S Bekhor - International journal of disaster …, 2018 - Elsevier

… place, is critical for the efficient **management** of the event. The usual … A large-scale **wildfire** took place in Haifa on November 24, … Shortly after the fire events, a web **survey** was developed …

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LANDFIRE: a nationally consistent vegetation, **wildland fire**, and fuel assessment

MG Rollins - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… of different **wildfire management** strategies, **wildfire** season … ; (3) communications and iterative **review** by local ecologists and … the probability of individual **wildfire spread** and severity and …

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A simple physical **model** for **forest fire spread**

E Koo, P Pagni, J Woycheese… - In: Gottuk, D.; …, 2005 - research.fs.usda.gov

… mechanisms of **forest fire spread** and introduced a simple physics-based fire **spread model** for … Pitts [12] provided a thorough **review** of wind effects on fire. Weise and Biging [13] built a …

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Modeling vorticity-driven **wildfire behavior** using near-field techniques

JJ Sharples, JE Hilton - Frontiers in Mechanical Engineering, 2020 - frontiersin.org

… An example **simulation** using the pyrogenic vector potential coupled to a **wildfire spread simulation** is shown in Figure 2 with s c = 0.5 and s w = 0.5 in Equation (8). These constants …

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Evacuation decision-making and **behavior** in wildfires: Past research, current challenges and a future research agenda

E Kuligowski - Fire Safety Journal, 2021 - Elsevier

… an understanding of human **behavior** in wildfires, which is an … decision-making and **behavior** during wildland-urban … support developments of evacuation **simulation** models, and …

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Describing wildland surface fuel loading for fire **management**: a **review** of approaches, methods and systems

RE Keane - International Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… models are being developed, it is critical that new fuel classification systems be built to balance ecological understanding of fuel **dynamics** with the new input **model** requirements to …

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Current **wildland fire** patterns and challenges in Europe: A synthesis of national perspectives

N Fernandez-Anez, A Krasovskiy… - Air, Soil and Water …, 2021 - journals.sagepub.com

… involved in **wildland fire** research and land **management**, … to assess and understand fire regime, **dynamics**, and post-fire … fire **behavior**, linkages between weather and **wildland fire** …

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How risk **management** can prevent future **wildfire** disasters in the wildland-urban interface

DE Calkin, JD Cohen, MA Finney… - Proceedings of the …, 2014 - pnas.org

… First, we **review** how the success, or lack thereof, of **wildfire** … fire **behavior** in the nearby wildlands, meaning that proper care of the HIZ separates home losses from **wildland fire behavior**, …

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Negative consequences of positive feedbacks in US **wildfire management**

DE Calkin, MP Thompson, MA Finney - Forest Ecosystems, 2015 - Springer

… this article we **review** the structure of US **wildfire management** … ecological drivers of **wildfire management** response in both … the **trajectory** of **wildfire** risk and associated **management** …

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Forest fire dynamics in Nepal: Regional trends and socio-ecological drivers

KP Joshi, S Giri, D Kuinkel, S Kuikel, R Devkota… - Trees, Forests and …, 2025 - Elsevier

… national **forest fire** analyses. This study investigates fire trends, key drivers, and public perceptions of **management** … **Survey** responses highlight strong community support for awareness …

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Detecting critical nodes in forest landscape networks to reduce **wildfire spread**

D Yemshanov, N Liu, DK Thompson, MA Parisien… - PloS one, 2021 - journals.plos.org

… to plan **wildfire** fuel treatments that minimize the risk of fire **spread** in … **simulation** modeling to estimate the probability of fire **spread** … A **review** of operational research studies in **forest fire** …

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Physical phenomena and length scales governing the behaviour of wildfires: a case for physical modelling

D Morvan - Fire technology, 2011 - Springer

… **review** concerning fire **spread** and fire behaviour models [9–11]. … as a “fully” physical **wildfire model** a **model** based on the solution of … For this **simulation** we found a rate of **spread** (ROS), …

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